Effects of Disturbance on Ecological Systems and Review of the Intermediate Disturbance Hypothesis

Abstract

Human disturbances on the environment have the potential to disrupt many ecological communities. For years, scientists have studied disturbances to ecological systems through various community metrics, and with the data proposed the intermediate disturbance hypothesis. This hypothesis states that at intermediate levels of disturbance, ecosystems experience highest levels of species richness, and has served as the basis for studying how communities would react to human disturbances. This study investigated how different levels of disturbance would affect various biotic and abiotic factors, measured as species richness, species diversity, and PAR levels, in Baton Rouge, Louisiana. Quadrat sampling was used in three areas: high disturbance area, low disturbance area, and medium disturbance area. Species types and abundance were recorded at each location, along with PAR levels, and data was collected in a Microsoft Excel spreadsheet. Data was imported into RStudio and analyzed using ANOVA testing, after which a Tukey HSD test was performed to determine significance between treatments. Results showed that there was a significant difference between level of disturbance to an area and species richness, as well as a significant difference between level of disturbance to an area and PAR levels. However, significance was not established between level of disturbance to an area and species diversity. Overall, these findings can demonstrate how human disturbances may influence communities and can allow scientists to predict shifts in ecological systems due to human disturbances.

Introduction

The interaction between humans and their environment has led to much discussion about the impacts of these interactions. Disturbances to ecological communities by humans has the potential to influence species stability, and even lead to entire collapse of the ecosystem (MacDougall et al., 2013). This may occur as species become unstable in these ecological systems through widespread effects that can ripple through all trophic levels as resources and food becomes scarce, leading to the ultimate decline of an entire community. Scientists have the potential to monitor these changes using community metrics. As ecosystems face changes caused by disturbances, we may use species richness and species diversity metrics to analyze how these changes are impacting local populations. Species richness allows us to quantify the number of different species in a location (Battisti et al. 2016), and while it may provide limited information about the well-being of a community, it can be used alongside other metrics to evaluate the dynamics of the area. Species diversity is another important community metric that looks at the richness of species in one location and analyzes how distributed they are in a count of ecological diversity (Battisti et al., 2016). These metrics together are thus the basis for how we can understand environmental disturbances, most notably through our understanding of the intermediate disturbance hypothesis (IDH). The IDH states that at a middle level of disturbance,

diversity within a community is at its highest due to top competitors being knocked out by the disturbances, allowing lower-level competitors to establish themselves in the community (Shea et al., 2004). Using the IDH, scientists can understand and even predict how ecological systems will react to human disturbances.

With the IDH as our framework, we can start looking at how communities are affected by both biotic and abiotic factors. Biotic factors are those that involve living organisms in the environment, which may include predator and prey interactions, as well as human interactions with this environment (Klanderud et al., 2015). Scientists can use community metrics, such as species richness and diversity, to characterize how well an area is doing because of these interactions. Abiotic factors involve non-living parts of the environment, such as wind speed, temperature, and soil conditions, and have been shown to impact how well a species can survive a stressful environment (Klanderud et al., 2015). One specific abiotic factor that is important to focus on is light availability, or photosynthetically active radiation (PAR). The productivity of plants is highly influenced by PAR, as the light is required for the plants to undergo photosynthesis, and too much or too little sunlight can put the community under stress (Lawlor, 1995). Human disturbances to ecological systems can occur either directly, such as through mowing an area of land, or indirectly, such as through increasing global temperatures via climate change. Therefore, analyzing communities that are facing human impact must be considered through both a biotic and abiotic lens.

Plants are important for studying the effects of disturbances on communities for several reasons. Firstly, plants serve as the foundation of the ecosystem by providing a food source for many animals and helping facilitate biodiversity in a successful community (Losapio et al., 2020). Because of this, plants are a good measure of how well an ecosystem is functioning and how changes to a community, such as through human disturbances, may influence the surrounding ecosystem. Additionally, plants are very responsive to changes in the ecosystem, whereby shifts in species richness and diversity would be notable through changes in the community structure by invasive species (Hobbs & Huenneke, 1992). Thus, human disturbances are easily observed in ecological systems by utilizing these community metrics.

Studying ecological systems using plants as the model organisms can allow researchers to study if the prediction put forth by the IDH holds true, and if diversity at intermediate disturbance levels yields highest species richness and diversity. Studying PAR levels may also demonstrate how abiotic factors affect community stability or is a result of biotic interactions with the environment. Understanding how disturbances can affect a community can thus allow scientists to predict how plant species will respond and influence the resulting community.

This experiment aims to answer these questions regarding the relationship between level of disturbance to a community and factors such as species richness, community diversity, and PAR levels. Our first hypothesis is that a medium level of disturbance to an area will have the highest level of species richness. We suspect this outcome because an intermediate level of

disturbance to an area will disrupt the top competitors, thus allowing more middle-tier competitors to exist, and therefore increasing the number of different species in that area. Our second hypothesis is that a medium level of disturbance will have the highest level of species diversity. We suspect this outcome following similar reasoning, being that intermediate disturbance will disrupt top competitors, which will allow for more middle-tier competitors to exist rather than one dominating top-competitor, thus increasing species diversity. Our final hypothesis is that an increase in disturbance to an area will cause in increase in PAR. We suspect this outcome because more disturbance to an area will result in the lack of trees and tall plants that provide shade, as compared to an area with low disturbance that will have more shade due to the presence of trees and tall plants. Understanding these relationships can provide insight into how human interaction with the environment may impact and disrupt community composition, which can allow scientists to predict how disturbances over time may impact the current ecosystem.

Methods

Data was collected from areas around Innovation Park Drive in Baton Rouge, Louisiana. The three treatment groups, which served as our independent variable, were areas of high disturbance, medium disturbance, and low disturbance. Quadrat sampling was utilized to isolate a one meter-squared area in each location and plant species were observed within this area using the iNaturalist application. Using this app, we took a clear image of each present plant species, identified the species names, counted the abundance of each species, and recorded data in a Microsoft Excel spreadsheet. To count extremely abundant species, the quadrat was divided into a tenth, the species was counted in just this region, and the obtained quantity was multiplied by 10. Four replicates were obtained from each location (high, medium, and low) to improve reliability of the results. Additionally, the abiotic factor of PAR was obtained at each site using a PAR sensor. Five measurements of PAR were collected and averaged for each of the four replicates at all three locations and was once again recorded in the Excel spreadsheet.

After data collection, community metrics equations were used to prepare the data for analysis. First, we calculated Shannon's diversity index using the equation: $H' = -\sum p_i(\ln(p_i)$, where H' is the Shannon's diversity index, p_i is the proportion of total abundance represented by that species, and ln is the natural logarithm. Using this equation, the H' values for each replicate were obtained as a measure of diversity. Once species richness, Shannon's diversity index, and average PAR for each replicate was obtained, data was analyzed using the RStudio program (version 2026.08.1+195). The data recorded in the Microsoft Excel spreadsheet was exported as a Comma-Separated Values (CSV) file, and imported into RStudio, along with the necessary code to perform three ANOVA tests, one for each dependent variable. Bar graphs were generated from our three ANOVA tests to analyze our hypotheses regarding the effect of varying levels of disturbance on species richness, Shannon's diversity index, and PAR. Once the bar graphs were produced, a Tukey's HSD test was performed for each ANOVA test to analyze the statistical significance between each treatment.

Results

We did find a statistical difference in the number of species, or species richness, between high, low, and medium disturbance level treatments. $F_{2,33} = 3.372$, $p = 0.0465$. The mean for high level of disturbance was 2.750000 ± 1.577505. The mean for low level of disturbance was 3.083333 ± 1.577505. The mean for medium level of disturbance was 4.333333 ± 1.577505. Tukey's HSD indicated that there was a significant difference in species richness for high levels of disturbance and medium levels of disturbance, in which a medium level of disturbance demonstrated increased species richness and a high level of disturbance demonstrated decreased species richness. Tukey's HSD also indicated that there was no significant difference in species richness between high and low levels of disturbance, as well as between low and medium levels of disturbance.

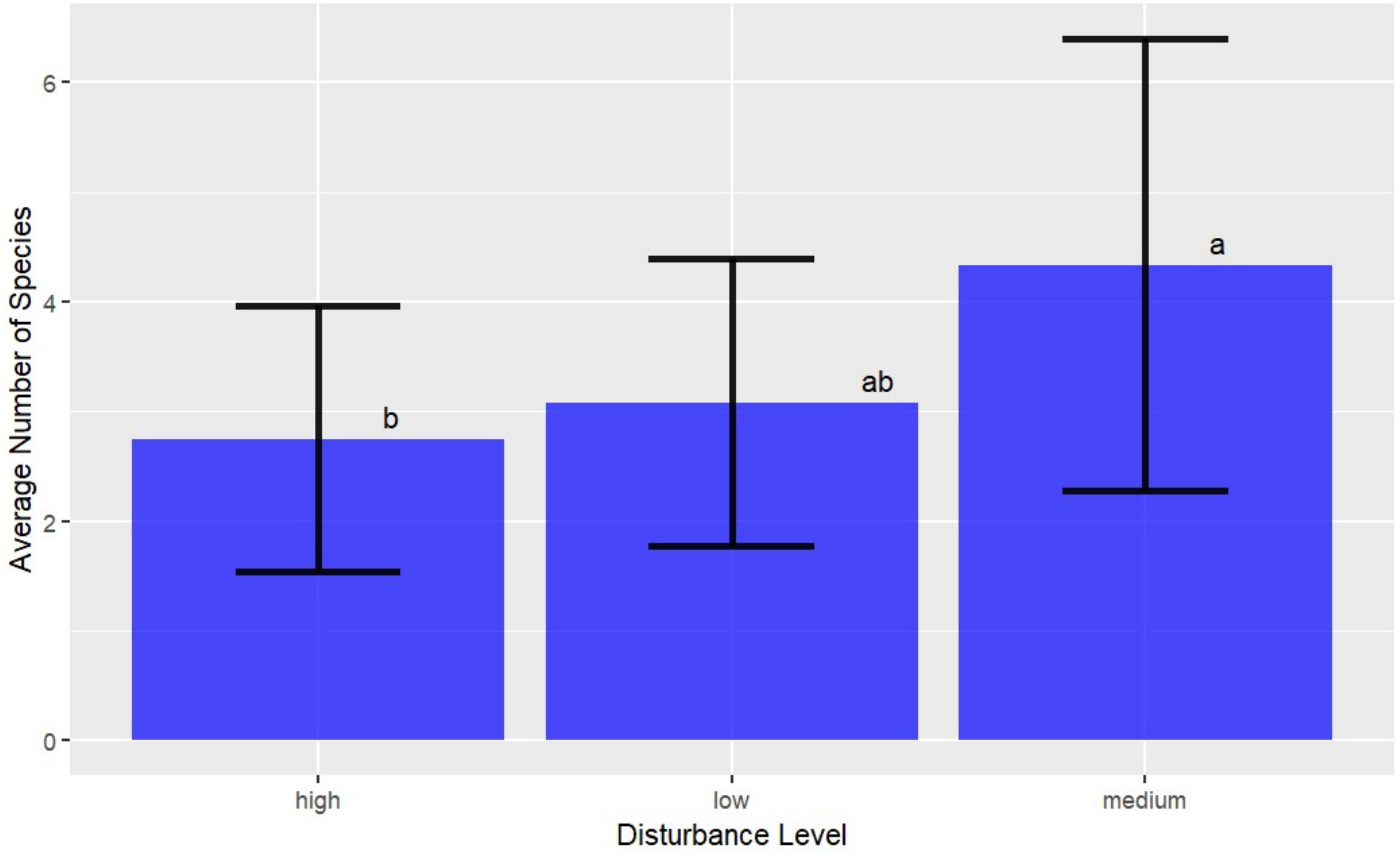


Figure 1. Average number of species (species richness) in locations of high, low, and medium levels of disturbance. The average number of species at the location of high disturbance level was 2.750000, of low disturbance level was 3.083333, and of medium disturbance level was 4.333333. The p-value was 0.0465 ($F_{2,33} = 3.372$). Error bars represent standard deviation. Tukey test was performed, where different letters indicate significant differences between treatments while letters in common indicate no significant difference.

We did not find a statistical difference in the Shannon's diversity index between high, medium, and low disturbance level treatments. $F_{2,33} = 2.863$, $p = 0.0713$. The mean for high level of disturbance was 0.4463710 ± 0.3979334. The mean for low level of disturbance was 0.8100134 ± 0.3979334. The mean for medium level of disturbance was 0.7455450 ± 0.3979334. Tukey's HSD indicated that there was no significant difference in Shannon's diversity index between high, low, or medium levels of disturbance.

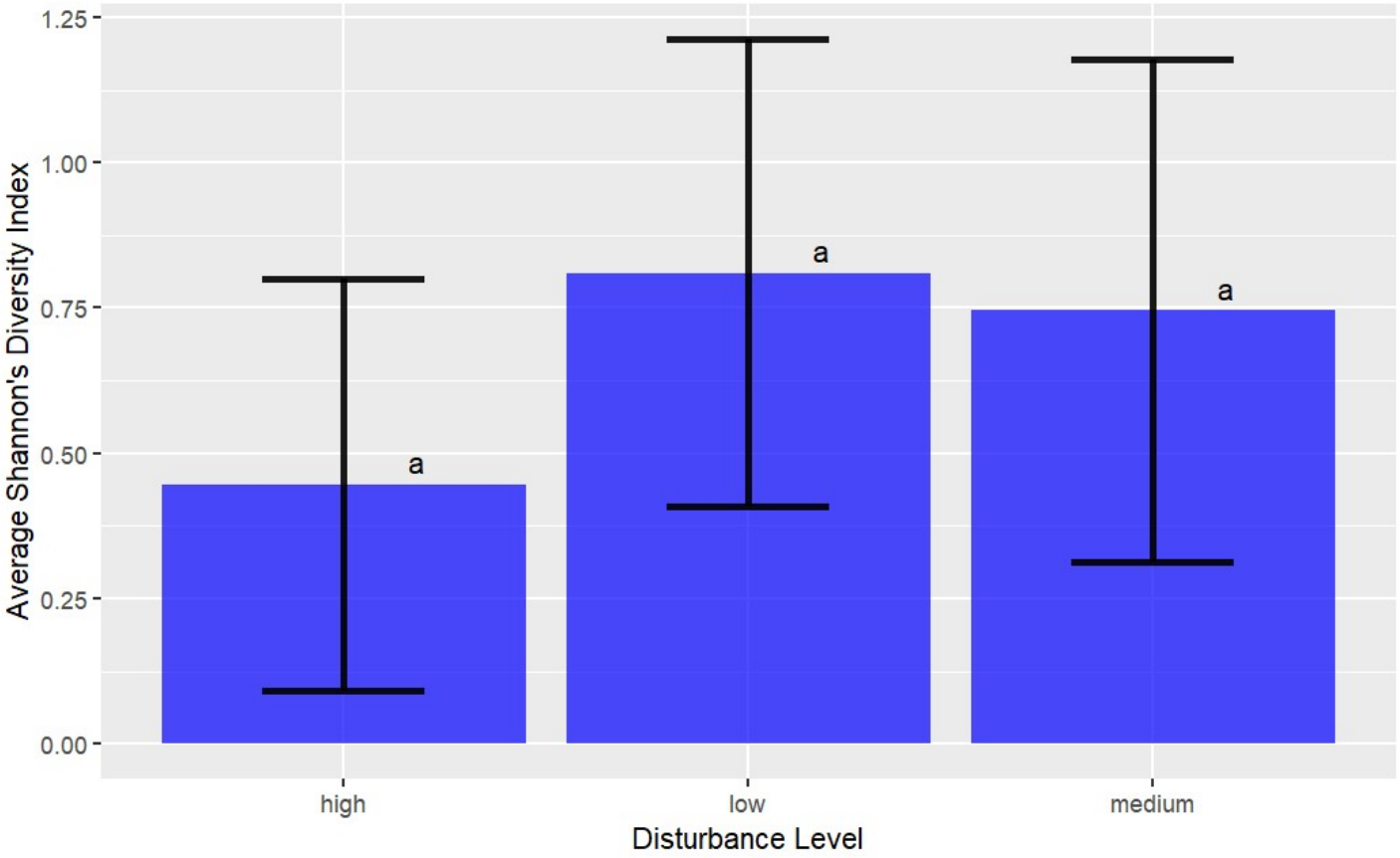


Figure 2. Average Shannon's diversity index in locations of high, low, and medium levels of disturbance. The average Shannon's diversity index at the location of high disturbance was 0.4463710, of low disturbance level was 0.8100134, and of medium disturbance level was 0.7455450. The p-value was 0.0713 ($F_{2,33} = 2.863$). Error bars represent standard deviation. Tukey test was performed, where letters in common indicate no significant difference between treatments.

We did find a statistical difference in PAR between high, medium, and low disturbance level treatments. $F_{2,33} = 31.96$, $p = 1.9e\text{-}08$. The mean for high level of disturbance was 886.3917 ± 249.6037. The mean for low level of disturbance was 174.7000 ± 249.6037. The mean for medium level of disturbance was 871.4167 ± 249.6037. Tukey's HSD indicated that there was a

significant difference in PAR between high and low levels of disturbance, as well as between medium and low levels of disturbance, in which both high and medium levels of disturbance demonstrated increased PAR and low levels of disturbance demonstrated decreased PAR. However, Tukey's HSD indicated no significant difference in PAR between high and medium levels of disturbance.

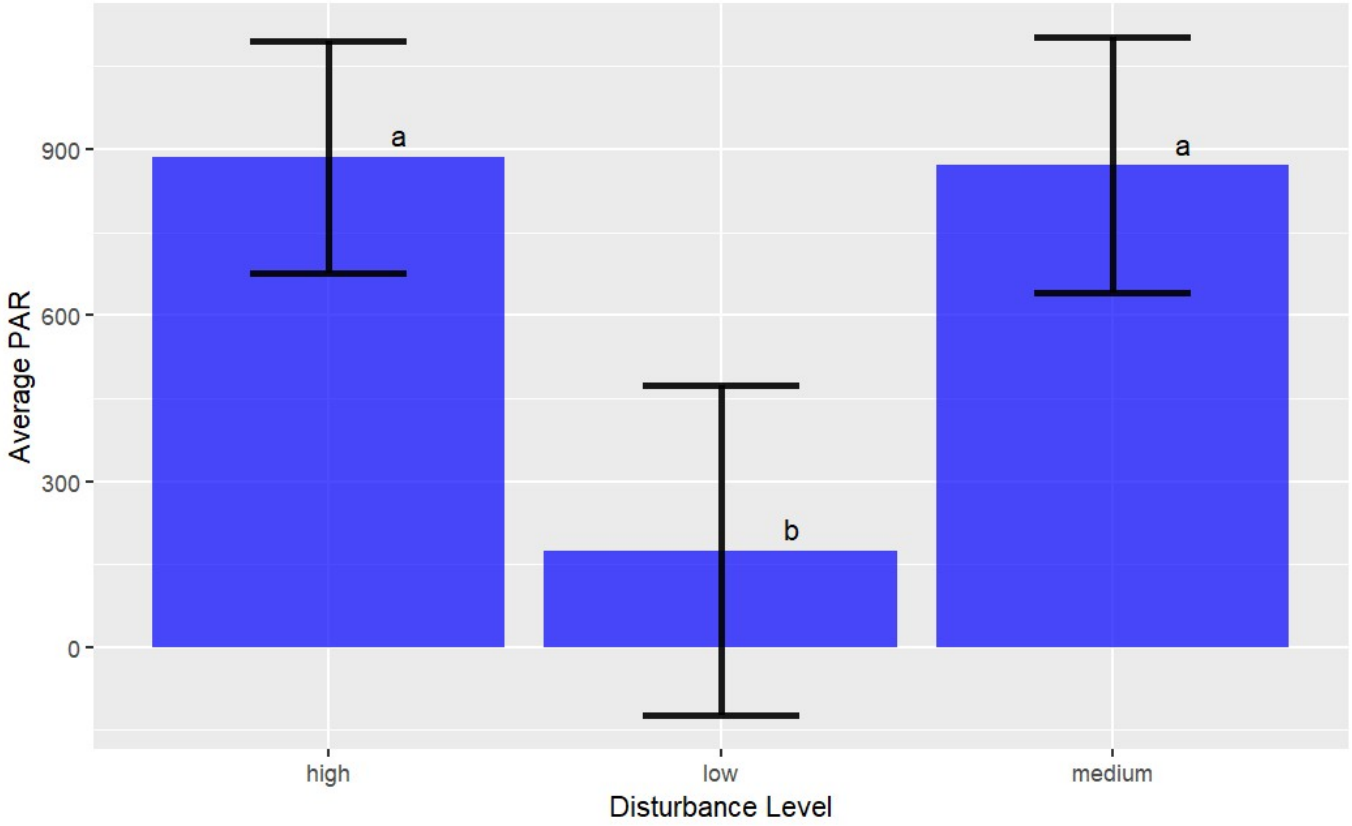


Figure 3. Average PAR in locations of high, low, and medium levels of disturbance. The average PAR at the location of high disturbance was 886.3917, of low disturbance level was 174.7000, and of medium disturbance level was 871.4167. The p-value was 1.9e-08 ($F_{2,33} = 31.96$). Error bars represent standard deviation. Tukey test was performed, where different letters indicate significant differences between treatments while letters in common indicate no significant difference.

Discussion

We rejected the null hypothesis stating that a difference in the amount of disturbance to an area will have no effect on the species richness and supported the alternative hypothesis that a difference in the amount of disturbance will have an effect on species richness. The biological meaning of these results ties into the prediction set forth by the IDH, which attributes higher species richness to a decrease in top-level competitors from a moderate amount of disturbance and allowing middle-level competitors to establish in the environment. These findings are

consistent with current research, seen by a study regarding the species richness in grasslands in which the authors found that the species richness of vascular plants in grasslands was at its highest at an intermediate level of grazing (Yuan et al., 2016). Thus, we conclude that our findings are relevant to the study of IDH, and present substantial evidence as to how levels of disturbance to an ecological system can influence species composition.

We failed to reject the null hypothesis that a difference in the amount of disturbance to an area will have no effect on species diversity. The biological meaning of these results indicates that there may be other environmental influences at play that could account for no significant difference in species diversity at different levels of disturbance. This perspective is reflected in the current literature, as there are various findings regarding the IDH and species richness. For example, data collected in a study of tropical rainforests found that at moderate disturbance levels, the community experienced greater species diversity as compared to other disturbance treatments (Molino & Sabatier, 2001). These findings directly support the predictions of the IDH and uphold the assumption that moderate levels of disturbance yield the highest species richness. However, another study looking at trampling disturbances in alpine grasslands found different results, in which there were no significant differences in species diversity across disturbance levels (Li et al., 2021). These findings, thus, do not support the predications put forth by the IDH and assumes that across levels of disturbance, there is a similar species diversity present. While our data is among many that challenges the IDH, if assuming the principles of the hypothesis to be true, we can assume our data is inconsistent due to sampling error, shown by a large calculated standard deviation. Samples measured in the medium disturbance area were collected in a very small range of the field that was located relatively close to the road, and results could have been improved if a larger part of the field was accounted for, especially if samples were taken further from the road.

We rejected the null hypothesis that a difference in the amount of disturbance to an area will have no effect on PAR and supported the alternative hypothesis that a difference in the amount of disturbance to an area will have an effect on PAR. The biological meaning of these results reflects how varying level of disturbances may impact the abiotic factors of a community and have further implications for the stability of the ecosystem. These findings are consistent with current literature, seen by a study regarding levels of disturbance on community structure, in which the authors found a negative relationship in the amount of disturbance to an area and the amount of light that come through the canopy (Haber et al., 2020). Thus, we can conclude that our findings on PAR levels and disturbance provide relevance to the discussion regarding how human disturbances can alter the structure and stability of a community by influencing abiotic factors.

Overall, the findings from this study suggest that disturbances, specifically human disturbances, can have an impact on community composition and stability. Future directions may look at improving statistical output, especially as to finding a relationship between disturbance levels and species diversity. In this research study particularly, taking samples further from an

active roadway may allow for a better representation of a moderate disturbance level community, as opposed to closer to the road, which may yield results more similar to a high disturbance level area. However, this study along with others in the field indicate through varying results that the focus in future studies should be directed towards analyzing other variables in the environment that could account for no difference in species diversity across disturbance levels. Thus, a better understanding of how these factors influence the community composition may produce answers as to how humans can minimize effects of disturbance levels on surrounding communities.

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